

SFTP Firebase Integration

Firestore integration (API subsystem)

Revision: 1 **Last modified:** 2026-07-11T00:00:00Z **Scope:** optional Firestore Admin SDK subsystem of the SFTP Enterprise management API (api/internal/firestore, ATM-007) + the web-app SDK config helper (scripts/firestore_config.sh, STREAM-6).

What this is

The management API can optionally initialize the **Firestore Admin SDK for Go** (firebase.google.com/go/v4) at startup. It is **OFF by default** and the API runs fully without it:

FIRESTORE_ENABLED	Behaviour
false (default)	API logs firestore: disabled and continues normally.
true, misconfigured	API fails fast with a clear error (missing project id, missing/unreadable/invalid service-account file) — no silent half-enabled state (§11.4.6).
true, configured	Admin SDK initializes from the service-account JSON; the API logs firestore: enabled (project <id>).

Crashlytics / Analytics / Performance hooks exist as **no-op-safe stubs**: the Go Admin SDK has no ingestion API for those surfaces (they are client-side mobile/web SDK surfaces), so the hooks log honestly instead of pretending to record telemetry. Client.Verify performs a real connectivity + credentials round-trip (Identity Toolkit GetUserByEmail probe; an authenticated “user not found”

answer is positive evidence of connectivity) — it requires the **Identity Toolkit API** enabled on the project, so it is not run automatically at startup.

Setup

Sources verified 2026-07-11 against current Firebase documentation: - <https://firebase.google.com/docs/admin/setup> (service account generation + Go init) - <https://firebase.google.com/docs/projects/learn-more> (project id location)

1. **Create / pick the Firebase project.** In the Firebase console create a project (or use an existing one). Note the **Project ID** from *Project settings* → *General*.
2. **Enable the surfaces you need** (per-product tabs in the console):
 - **Crashlytics:** *Run* → *Crashlytics* — connect an app that reports crashes (mobile/web clients own this; the API does not ingest).
 - **Analytics:** enabled automatically when you add an app with Google Analytics.
 - **Performance Monitoring:** *Run* → *Performance*.
 - **Identity Toolkit** (only if you intend to call `Client.Verify` / the health probe): enable the *Identity Platform* / *Identity Toolkit API* in the Google Cloud console for the project.
3. **Generate the service-account key.** *Project settings* → *Service accounts* → *Generate new private key* → *Generate key*. This downloads a JSON file. Treat it as a high-value secret (Firebase's own guidance).
4. **Store the key outside git.** Move it to the git-ignored location:

```
mkdir -p secrets && chmod 700 secrets
mv ~/Downloads/<project>-firebase-adminsdk-*.json secrets/
    firebase-service-account.json
chmod 600 secrets/firebase-service-account.json
```

`service-account*.json` / `firebase-service-account*.json` and `secrets/` are git-ignored (§11.4.10/.30). **NEVER commit a real service account.**

5. **Set env (in the git-ignored .env, template: .env.example):**

```
FIREBASE_ENABLED=true
FIREBASE_PROJECT_ID=<your-project-id>
```

```
FIREBASE_SERVICE_ACCOUNT_PATH=./secrets/firebase-service-  
account.json  
FIREBASE_WEB_APP_ID=<1:PROJECT_NUMBER:web:HASH> # only for  
step 6
```

6. (Optional) Fetch the web SDK config for the admin SPA:

```
firebase login # once  
firebase apps:list # find the WEB app id if unsure  
scripts/firebase_config.sh --check # or plain run to  
(re)fetch
```

This writes `web/src/firebase-config.json` (git-ignored, never printed — see `docs/scripts/firebase_config.md`).

7. Start the API. With `FIREBASE_ENABLED=true` the log line `firebase: enabled (project <id>)` confirms initialization; with `false`, `firebase: disabled` confirms the subsystem is inert.

Tests

```
cd api && go test ./internal/firebase/...
```

Covers: disabled path is inert + logs `firebase: disabled`; each misconfigured-enabled path (no project id, no key path, missing file, directory instead of file, malformed JSON, undecryptable private key) fails fast with a clear error. The tests exercise the real Admin-SDK init logic (`firebase.NewApp + option.WithCredentialsFile`) — no tautologies.

Cross-references

- `api/internal/firebase/firebase.go` — subsystem source (contract in the package doc comment).
- `config_schemas/firebase.yaml` — field schema + JSON Schema.
- `scripts/firebase_config.sh` + `docs/scripts/firebase_config.md` — web SDK config fetch.
- `.env.example` — env template; constitution §11.4.10 (credentials), §11.4.30 (git-ignore discipline), §11.4.99 (latest-source verification).